Note:Students have to write programs along with atleast 2 outputs everylab and get the observation signed by the faculty incharge. For every lab Observation carries Marks

Solutions

1) Sum of Natural Numbers

```
n = int(input("Enter a number: "))

sum = 0
i = 1

while i <= n:
    sum = sum + i
    i = i + 1

print("Sum =", sum)
```

Sample input and output

Enter a number: 5
Sum = 15

2) Check Positive, Negative or Zero

```
n = int(input("Enter a number: "))

if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

3) Find the Largest of Three Numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    largest = a
elif b >= a and b >= c:
    largest = b
else:
    largest = c

print("Largest number =", largest)
```

4) Check Even or Odd

```
n = int(input("Enter a number: "))

if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

5) Reverse a Number

```
n = int(input("Enter a number: "))

reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10
```

```
print("Reversed number =", reverse)
```

6) Palindrome Number Check

```
n = int(input("Enter a number: "))

original = n
reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10

if original == reverse:
    print("Palindrome number")
else:
    print("Not a palindrome number")
```

7) Count Vowels in a String

```
s = input("Enter a string: ")

count = 0

for ch in s:
    if ch in "aeiouAEIOU":
        count += 1

print("Number of vowels:", count)
```

8) Find the Largest Digit

```
n = int(input("Enter a number: "))

largest = 0

while n > 0:
    digit = n % 10

    if digit > largest:
        largest = digit

    n = n // 10

print("Largest digit =", largest)
```

9) Find the Smallest Digit

```
n = int(input("Enter a number: "))

smallest = 9

while n > 0:
    digit = n % 10

    if digit < smallest:
        smallest = digit

    n = n // 10
```

```
print("Smallest digit =", smallest)
```

10) Count Words in a Sentence

```
s = input("Enter a sentence: ")

sentence = input("Enter a sentence: ")

count = 0
in_word = False

for ch in sentence:
    if ch != " " and not in_word:
        count += 1
        in_word = True
    elif ch == " ":
        in_word = False

print("Number of words:", count)
```

Prepared by,
Anil s